BIO 004 · HUMAN ANATOMY · SOLANO COMMUNITY COLLEGE

Pre-work · Integumentary System

Integumentary System. Read the Part A chapter first, then work this in order. This is the first organ where all four tissue types show up together, so it is where the module gets used rather than listed.

Module 1 · 4 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the integumentary chapter into mini visuals. Seven chunks, and the ladder is the one you will use on every exam.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 LADDER

The five strata

Five rungs, basale at the bottom. Beside each rung write what the keratinocyte is doing at that level. The mnemonic goes in the margin, not on the ladder.

2 HUB

The four cells of the epidermis

One hub, four spokes: keratinocyte, melanocyte, intraepidermal macrophage, Merkel cell. Percentage on one line, job on the next. Mark which stratum each lives in.

3 CHAIN

Melanin transfer

Melanocyte, arrow, projection between keratinocytes, arrow, granules enter the keratinocyte, arrow, granules cluster over the nucleus. Draw the last step, do not just write it. That veil is the whole point.

4 SPLIT

Papillary against reticular dermis

One split. Fraction of the dermis, fibers, and what lives in each on the branches. Add the three structures inside a dermal papilla.

5 GRID

The four skin glands

Sebaceous, eccrine, apocrine, ceruminous down the side. Where they are, what they secrete, where the duct opens, and when they switch on across the top.

6 HUB

The nail

Nail at the center. Spoke to body, free edge, root, lunula, hyponychium, eponychium, matrix. One line each on what it is or does.

7 CHAIN

Thermoregulation, both directions

Two chains from one starting point. Too warm goes one way, too cold the other. Put vessels and sweat on both chains and shivering on one.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The skin in cross section

BRAIN DUMP FIRST

Two minutes. Every layer and every structure in the skin.

Draw the full thickness: epidermis, dermis with papillary and reticular regions, and hypodermis. Add a hair follicle running down, a sebaceous gland on it, an arrector pili muscle, an eccrine gland with its duct reaching the surface, dermal papillae with capillary loops and Meissner corpuscles, and a lamellated corpuscle in the hypodermis. Label everything.

In your notebook, head the page **M1-10 The skin in cross section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can draw a horizontal line across your own drawing at any depth and say what a burn to that depth would destroy.

DRAWING 2

The five strata, cell by cell

BRAIN DUMP FIRST

One minute. The strata in order and what changes between them.

Zoom in on the epidermis alone. Draw the five strata stacked, with the cells changing shape as they rise: cuboidal and dividing at the base, spiny in the middle, granular above, clear, then flat dead squames at the top. Mark where the nucleus disappears and where lamellar granules release their sealant. Mark which layer is missing in thin skin.

In your notebook, head the page **M1-11 The five strata, cell by cell** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain why the epidermis survives without a blood supply, using only your drawing.

DRAWING 3

Hair follicle and bulb

BRAIN DUMP FIRST

One minute. The parts of a hair and everything attached to the follicle.

Draw one follicle from the surface down to the bulb. Show the shaft above the skin and the root below, the three concentric layers, the external and internal root sheaths, the dermal root sheath, and at the base the bulb with the papilla and the matrix. Attach the sebaceous gland, the arrector pili, and the hair root plexus.

In your notebook, head the page **M1–12 Hair follicle and bulb** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say where the hair actually grows from and why cutting it does not change that.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A burn destroys the epidermis and the upper dermis but leaves hair follicles and glands intact. Predict whether it regenerates, and say why.

Answer. Yes, it regenerates, from the follicles and glands.

Reasoning. Ask one question of every burn: is there a source of new epidermal cells left inside the wound. The stratum basale is gone at the surface, but hair follicles and glands are lined with epidermal cells and they sit deep in the dermis, so they survive. New epithelium spreads outward from each surviving structure until the sheets meet. A full-thickness burn destroys those too, which is why it cannot heal from the middle and needs grafting.

Set A · Layers and cells

1. Name the strata in order, deep to superficial, for thick skin.

   ____

2. Which stratum is missing in thin skin, and where on the body would you find thick skin?

   ____

3. The cell that alerts the immune system to an invader in the epidermis is the ____________________.
4. Which epidermal cell is least numerous, where does it sit, and what does it contact?

   ____

Set B · Color and pigment

1. Two people have very different skin color but the same number of melanocytes. Explain the difference.

   ____

   ____

   ____

2. A patient appears yellow in the skin and the sclera. Name the finding, the pigment, and the organ you would suspect.

   ____

   ____

   ____

3. A patient with dark skin is suspected of being cyanotic. Say where you would look instead, and why.

   ____

   ____

   ____

4. Explain why a tan is protective and why melanocytes are still damaged by the same exposure.

   ____

   ____

   ____

Set C · Structures and consequences

1. A gland opens onto a hair follicle in the axilla and switches on at puberty. Name it. ____________________
2. Damage to the nail matrix. Predict the effect on the nail and say why.

   ____

   ____

   ____

3. A patient loses the ability to sweat over a large area. Predict the consequence in hot weather and name the function lost.

   ____

   ____

   ____

4. Explain why a steroid cream works on inflamed skin but a large protein would not cross at all.

   ____

   ____

   ____

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M1-1

The five strata

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

M1-2

The four cells of the epidermis

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M1-3

Melanin transfer

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

M1-4

Papillary against reticular dermis

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M1-5

The four skin glands

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M1-6

The nail

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M1-7

Thermoregulation, both directions

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.